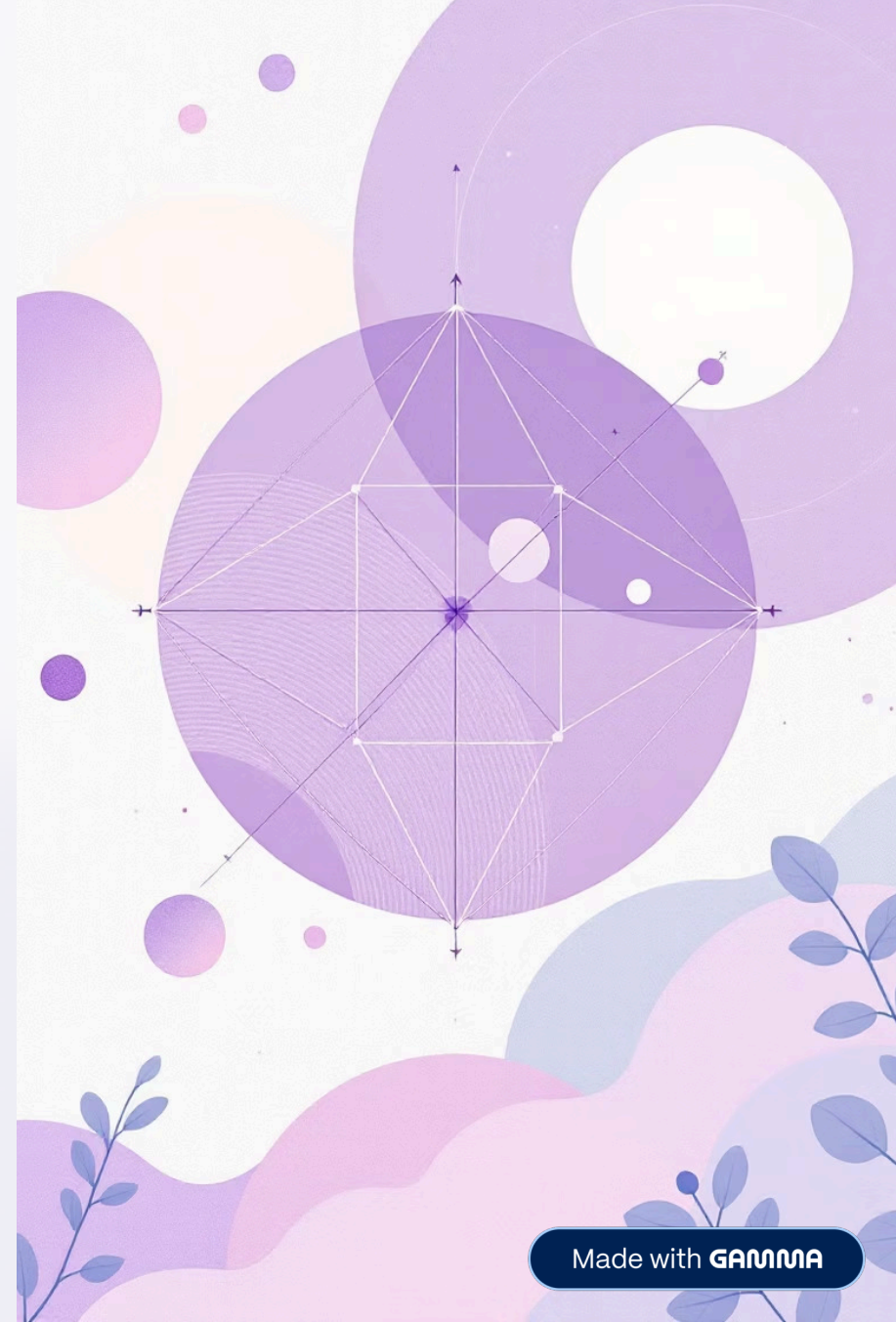


Eigenvalues and Eigenvectors

Eigenvalues and eigenvectors are fundamental concepts in linear algebra derived from the German word 'eigen' meaning proper or characteristic. These mathematical tools are essential for solving systems of differential equations, analyzing population growth models, and have critical applications in quantum mechanics and economics.



Core Definition

The Fundamental Equation

A non-zero vector $\mathbf{x}$ is an eigenvector of square matrix A if there exists a number λ such that:

$$A\mathbf{x} = \lambda\mathbf{x}$$

where $A = [a_{ij}]_{n \times n}$ is an n -rowed square matrix and λ is the eigenvalue.

Geometric Interpretation

The eigenvalue λ determines how the linear operator compresses or stretches the eigenvector $\mathbf{x}$. If λ is negative, the direction of the eigenvector reverses.



Characteristic Equation

The characteristic equation is fundamental to finding eigenvalues:

$$\det(A - \lambda I) = 0$$

01

Form Characteristic Matrix

Construct $(A - \lambda I)$ where I is the identity matrix

03

Solve for Eigenvalues

Set determinant equal to zero and solve for λ

02

Calculate Determinant

Find $\det(A - \lambda I)$, yielding a polynomial in λ of degree n

04

Find Eigenvectors

Substitute each λ back into $(A - \lambda I)x = 0$ and solve

Quick Calculation Methods

2×2 Matrices

Characteristic equation:

$$\lambda^2 - S_1\lambda + S_2 = 0$$

where S_1 = sum of diagonal elements and S_2 = determinant of A

3×3 Matrices

Characteristic equation:

$$\lambda^3 - S_1\lambda^2 + S_2\lambda - S_3 = 0$$

where S_1 = sum of diagonal elements, S_2 = sum of minors of diagonal elements, S_3 = determinant of A

📌 **Key Properties:** The sum of eigenvalues equals the trace (sum of diagonal elements), and the product of eigenvalues equals the determinant.

Nature of Eigenvalues for Special Matrices

1

Triangular Matrix

Eigenvalues are the diagonal elements

2

Real Symmetric Matrix

Eigenvalues are real numbers

3

Skew Symmetric Matrix

Eigenvalues are purely imaginary or zero

4

Hermitian Matrix

Eigenvalues are real numbers

5

Skew Hermitian Matrix

Eigenvalues are purely imaginary or zero

6

Unitary Matrix

Eigenvalues have unit modulus

7

Orthogonal Matrix

Eigenvalues have unit modulus



Key Theorems on Eigenvector Relationships

Linear Independence

- If $\mathbf{x}$ is an eigenvector, then $k\mathbf{x}$ is also an eigenvector for the same eigenvalue ($k \neq 0$)
- An eigenvector cannot correspond to more than one eigenvalue
- Eigenvectors for distinct eigenvalues are linearly independent
- For symmetric matrices, eigenvectors for distinct eigenvalues are orthogonal
- Eigenvectors for equal eigenvalues may be linearly independent or linearly dependent.

Matrix Operations

- If λ is an eigenvalue of A , then $1/\lambda$ is an eigenvalue of A^{-1}
- If λ is an eigenvalue of A , then λ^k is an eigenvalue of A^k
- If λ is an eigenvalue of A , then $\lambda \pm k$ is an eigenvalue of $A \pm kI$
- A and A^T have the same eigenvalues but may have different eigenvectors

Worked Example: Finding Eigenvalues

Given matrix A, find eigenvalues for various transformations:

$$A = \begin{bmatrix} 3 & -1 & 1 \\ -1 & 5 & -1 \\ 1 & -1 & 3 \end{bmatrix}$$

Using the characteristic equation $\lambda^3 - S_1\lambda^2 + S_2\lambda - S_3 = 0$:

11

S_1 (Trace)

Sum of diagonal: $3 + 5 + 3$

36

S_2 (Minors)

Sum of minors of diagonal elements

36

S_3 (Determinant)

Determinant of matrix A

This yields eigenvalues: $\lambda = 2, 3, 6$



A: 2, 3, 6



A^{-1} : 1/2, 1/3, 1/6



A^2 : 4, 9, 36



$A^3 + 2I$: 10, 29, 218

Finding Eigenvectors: Step-by-Step Process

For each eigenvalue λ , solve $(A - \lambda I)\mathbf{x} = \mathbf{0}$ using Cramer's rule:



Substitute λ

Form the system $(A - \lambda I)\mathbf{x} = \mathbf{0}$

$$\frac{f}{dx}$$

Apply Cramer's Rule

Find ratios $x:y:z$ using determinants



Express Solution

Write eigenvector as $\mathbf{x} = t\mathbf{x}_1$ where t is a parameter

Example: For $\lambda = -1$ in a 3×3 system, if we get $x/-6 = y/-2 = z/7 = t$, then the eigenvector is:

$$\mathbf{x} = t \begin{bmatrix} -6 \\ -2 \\ 7 \end{bmatrix}$$

□ The vector $\mathbf{x}_1 = [-6, -2, 7]^T$ forms a basis for the eigenspace corresponding to $\lambda = -1$.

Algebraic vs. Geometric Multiplicity

Algebraic Multiplicity

The number of times an eigenvalue λ appears as a root of the characteristic equation $\det(A - \lambda I) = 0$.

Example: If $\lambda^3 - 6\lambda^2 + 11\lambda - 6 = 0$ gives $\lambda = 2, 2, 2$, then $\lambda = 2$ has algebraic multiplicity 3.

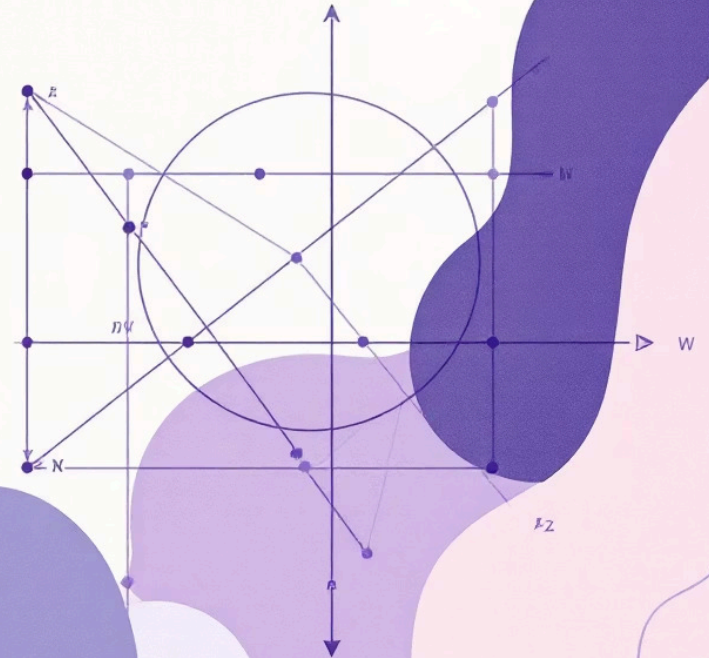
For a repeated eigenvalue, geometric multiplicity $\leq$ algebraic multiplicity. When they're equal, the matrix has a complete set of eigenvectors.

Geometric Multiplicity

The number of linearly independent eigenvectors corresponding to an eigenvalue, calculated as:

$$\text{Geometric multiplicity} = n - \text{rank}(A - \lambda I)$$

where n is the number of unknowns.



Applications and Practice

Differential Equations

Eigenvalues determine solution behavior and stability of systems

Population Models

Dominant eigenvalue indicates long-term growth rate

Quantum Mechanics

Eigenvalues represent observable quantities like energy levels

Key Takeaways

- Eigenvalues and eigenvectors reveal fundamental properties of linear transformations
- The characteristic equation $\det(A - \lambda I) = 0$ is the gateway to finding eigenvalues
- Special matrices have predictable eigenvalue properties (real, imaginary, unit modulus)
- Understanding multiplicity helps determine the complete eigenspace structure
- These concepts are essential tools across mathematics, physics, and engineering